



MARC
Management

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1.0 Introduction

This Best Practice is aimed at restoring liners on Ultra Class trucks using Plateau Honing.

Scope

C175 and 3500 liners from Finish Hone to Plateau Honing, considering the reusability tolerances of the liner in terms of honing angle, roughness and internal diameter.

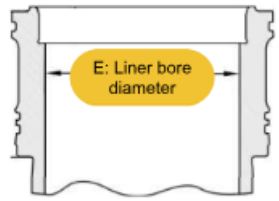
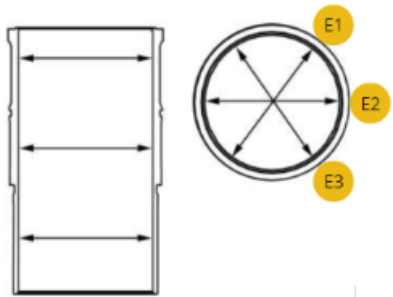
2.0 Best Practice Description

This Best Practice details the procedure and standards for performing liner honing on C175 and 3500 engines. It provides a comprehensive guide from the necessary preliminary steps prior to honing through to acceptance criteria for dimensional evaluation, visual inspection and roughness measurement after the process. This document also sets out precise protocols to ensure quality and precision at every stage of the process, ensuring that honed liners meet the required standards to guarantee optimum performance.

3.0 Implementation Steps

- Preliminary steps (carried out in the CRC workshop)

Acceptance criteria for dimensional assessment according to GSP-STT-VB-0003.

		EXTENDED REUSABILITY IRON TEST FOR INTERNAL DIAMETER OF C175 AND 3500 ENGINE LINERS - TIF		GSP-STT-VB-0003 Version 02 Date: 08-09-23
			Models C175 and 3500	
EVALUATION				Task to be performed
Model	Specs	ø max in good conditions *	ø max with stains, scratches or pitting **	Diam.
C175	175.025+/-0.025	175.065	175.040	E:
3500	170.025+/-0.025	170.065	170.040	E:
				CRC performs Flex honing. * Not necessary to send to MSH ** Must be send to MSH for R+J+H Note: If ø is out-of-measure, then it goes out-of-service
Elaborado por: Omar Soto / Aprobado por: Miguel Mauriolo				División de Soporte al Producto

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Recovery of C175 and 3500 Liners using Plateau Honing

DATE
12/20/2024

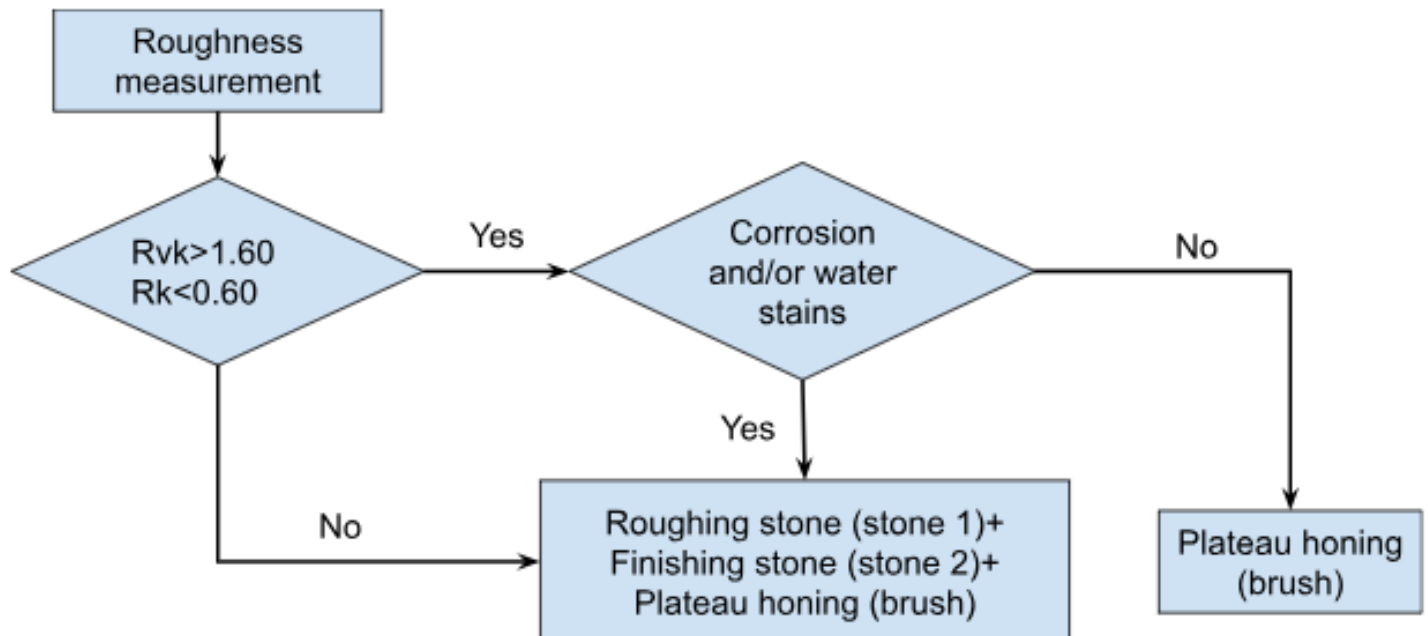
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- Acceptance criteria for visual inspection. The following will render the cylinder liners unusable: Stains (see reusability guide for more details), scratches, pitting and out-of-size.



- Acceptance criteria for roughness measurement.



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1. Parameters for Honing in Rottler H85A

- Sensitivity: 6.00
- Choice of stones: [514-9-14R] + [514-9-14J]
- Brush selection: [514-9-14H]
- **Roughing stone** (stone 1) operating ranges:
 - 514-9-14R
 - 50% rough load
 - 40% finish load
 - 10% plateau load
 - 180 RPM.
 - 0.020 mm
- **Finishing stone** (stone 2) operating ranges:
 - 514-9-14J
 - 30% rough load
 - 20% finish load
 - 10% plateau load
 - 180 RPM
 - 0.005 mm
- **Plateau honing** (brushes) operating ranges:
 - 514-9-14H
 - 30% rough load
 - 20% finish load
 - 8% plateau load
 - 4 strokes
 - 80 RPM.
- Setting angle: 40°
- Stone:
 - Upper and lower height margin: 22.00 mm

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Honing

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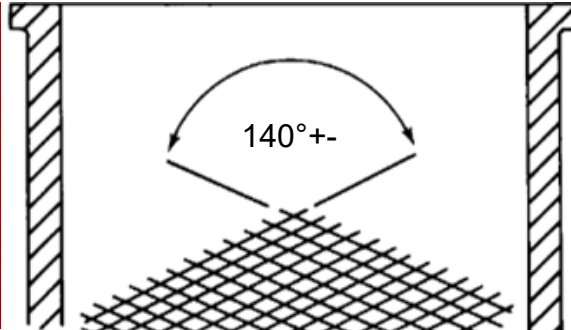
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2. Evaluation after honing

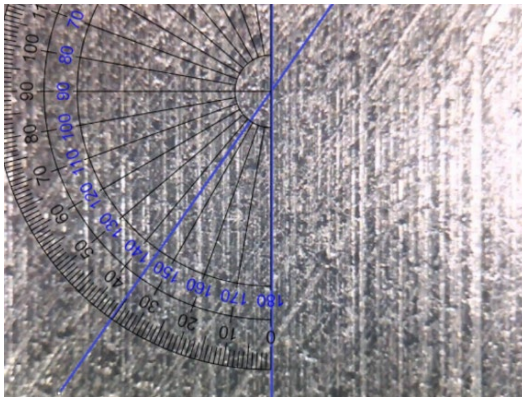
- Image taken after honing. The honing angle is $140^\circ \pm 6^\circ$.



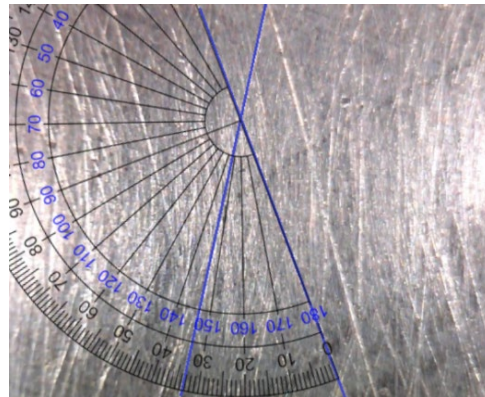
Finish after honing



Honing tolerance

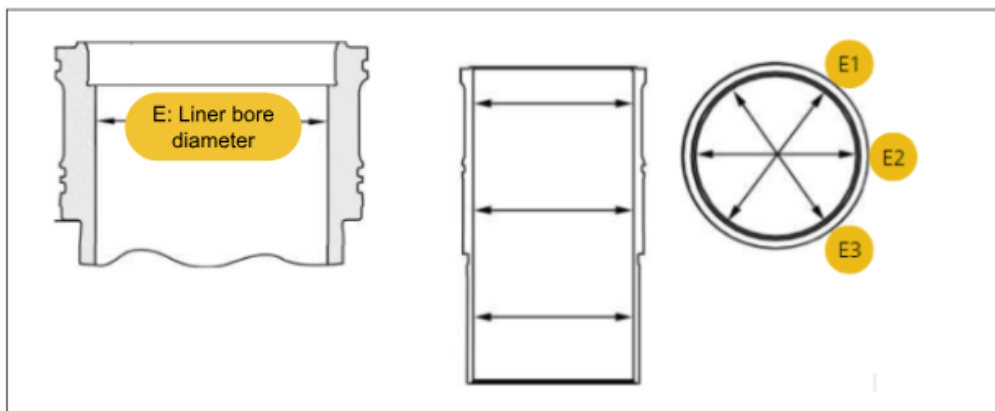


New Cylinder Liner: 145°
[C175]



New Cylinder Liner: 146°
[3500]

- The dimension of the housing (E) must be within the following limits:
 - [C175]: 175.000 - 175.065 mm.
 - [3516]: 170.000 - 170.065 mm.



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- Roughness
 - Cut off (λ_c) = 2.5mm.
 - R_{vk} = 1.2 μm - 3.8 μm .
 - R_k = 1.6 μm (max).
 - R_{pk} = 0.6 μm (max).
 - $Mr2\%$ = 80% + / - 10%.

4.0 Benefits

- It provides a clear and standardized framework for performing C175 and 3500 liner honing, ensuring quality in the process.
- It helps to optimize the use of resources such as time, labor and tools, resulting in greater efficiency and cost reduction, with annual savings of:
 - \$386 per C175 liner recovered.
 - \$276 per 3500 liner recovered.

5.0 Resources Required

- Initial finishing accessories kit
 - 4 Stone holders: 514-9-6R
 - 1 Head: 514-9-4A
 - 1 Shaft: 514-9R
 - 4 Stones: 514-9-14R
- Final finishing accessories kit
 - 4 Stone holders: 514-9-6R
 - 1 Head: 514-9-4A
 - 1 Shaft: 514-9R
 - 4 Stones: 514-9-14J
- Plateau honing accessories kit
 - 4 Stone holders: 514-9-6R
 - 1 Head: 514-9-4A

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- 1 Shaft: 514-9R
- 4 Stones: 514-9-14H

ITEM	PROCESS	HOLDER STONE	TOOL	QTY	CODE
1	ROUGHING PROCESS	514-9-6R	STONE	4	514-9-14R
2	FINISHING PROCESS		STONE	4	514-9-14J
3	PLATEAU PROCESS		BRUSH	4	514-9-14H



Stone 514-9-14R



Holder Stone 514-9-6R

6.0 Supporting Attachments / References

- [SEBF8068-36, "Inspection and Reuse Guideline of Cylinder Liners in Cat Engines"](#)
- [Blueprint, "Camisa C175"](#)
- [Blueprint, "Camisa 3500"](#)

7.0 Related Best Practices

None

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8.0 Acknowledgements

This Best Practice was written by:

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